SELECTED PROBLEMS ON CIRCUIT THEORY (EE-212).

1. Reduce the network of fig. 1 to a single voltage source with a series resistance at terminal A-B.

[Ans. 10 V, 4 Ω]

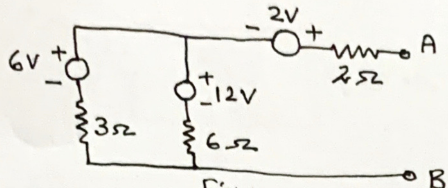


Fig. 1

2. The impedance across the Terminal A-B as shown in fig. 2 is to be a pure resistance of 500 Ω at $f = \frac{10^5}{2\pi}$ Hz. Determine L and C.

[L = 2 mH ; C = 0.01 μ

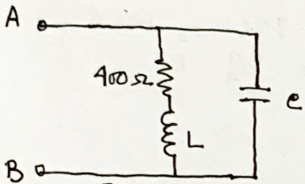


Fig. 2

3. Find out the conditions under which the power factor of the circuit shown in fig. 3 will be unity at all frequency.

[Ans. $R_1 = R_2$ and $L = CR_1^2$]

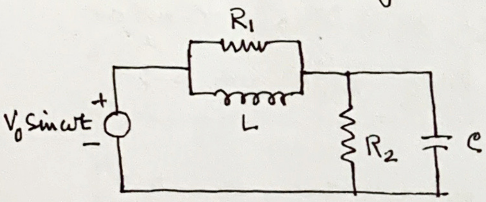


Fig. 3

4. Convert the current sources into equivallent voltage sources in fig. 4 and find out V_0.

[Ans. 2 Volt]

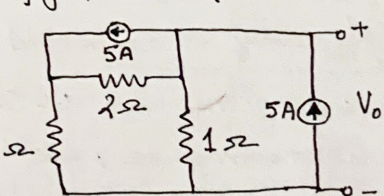


Fig. 4

5. A parallel RLC circuit is required to have resonance at 1000 rad/sec. The impedance at resonance is 10^5 ohm and the bandwidth is 100 rad. Determine R, L and C values.

[Ans. R = 990 Ω, L = 9.9 H
C = 0.1 μF]

6) Find out the resonant frequency of the circuit shown in fig. 5.
If $v(t) = 2 \sin t$ volt then find out $i(t)$.

[Ans. 1 rad/sec, $i(t) = 1.6 \sin t$ A.

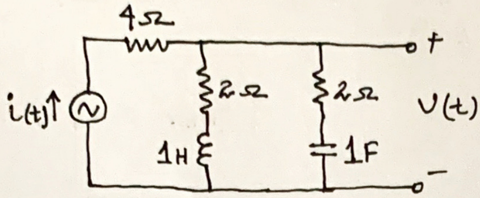


Fig. 5

7) Find out the resonant frequency, 3dB bandwidth and Q-factor at resonance of the circuit shown in fig. 6.

[Ans. $f_r = 796$ Hz, BW = 39.8 Hz, $Q = 20$]

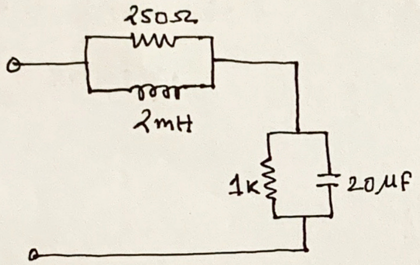


Fig. 6

8) A coil of inductance L and resistance R is in shunt with a capacitor C. The driving point impedance $Z(s)$ has a zero at $s = -\frac{1}{2}$ and poles at $\left(-\frac{1}{4} \pm j \frac{\sqrt{15}}{4}\right)$. Also $Z(j0) = 1\ \Omega$. Calculate the values of R, L and C.

[Ans. $R = 1\ \Omega$, $L = 2$ H, $C = \frac{1}{2}$ F].

[Hint. X_L of inductor $= j\omega L = sL$ and $X_c = \frac{1}{j\omega c} = \frac{1}{sc}$].
$Z(s)$ is the impedance represented in s-domain.

9) A 100 volt (peak value) source with variable frequency is connected to a series RLC circuit with $R = 50\ \Omega$, $L = 0.5$ H, and $C = 50\ \mu F$. Determine the resonant frequency, current at resonance, and Q-factor.

[Ans. 31.83 Hz, 2 Amp., $Q = 2$]

10) Determine the frequency at which the voltage V_o shown in fig. 7 will be zero.

[Ans. 2.25 KHz].

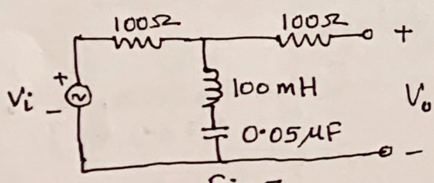


Fig. 7

Transform the voltage sources of fig. 8 into equivallent current sources and find out the current through R_L.

[Ans. Current through R_L is 1

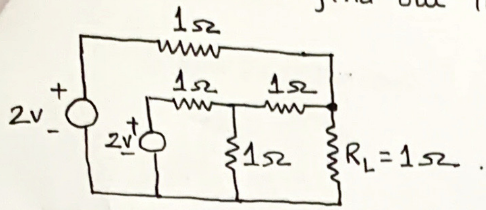


Fig. 8

(12) Find out the voltage V_{AB} of the fig. 9 network.

[Ans. $V_{AB} = 75.2 \angle 55.2^\circ$ V]

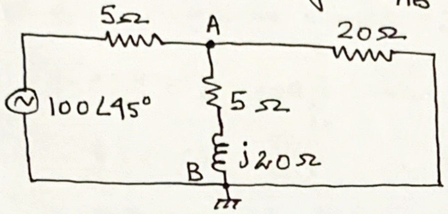


Fig. 9

(13) Calculate the current 'I' in the circuit shown in fig. 10.

[Ans. 1 Amp]

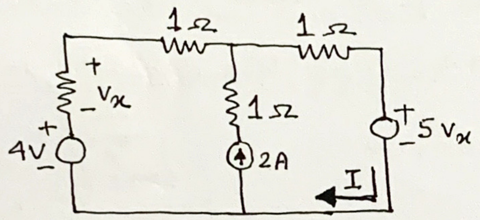


Fig. 10

(14) Determine the value of source voltage of the fig. 11.

[Ans. V = 4.04 volt].

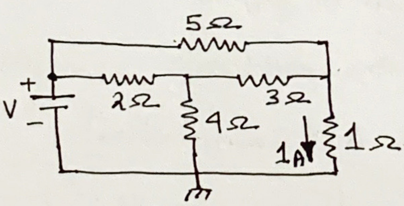


Fig. 11

(15) Find out the voltage v_x of the circuit shown in fig. 12

[Ans. $v_x = \frac{4}{9}$ Volt]

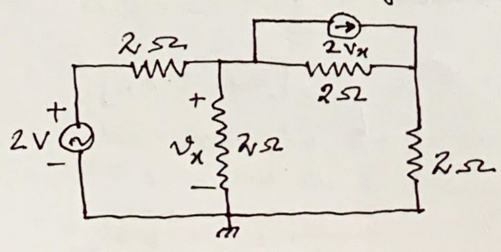


Fig. 12

(6) Find out the power delivered to the dependent voltage source shown in fig. 13.

[Ans. −2.337 watt]

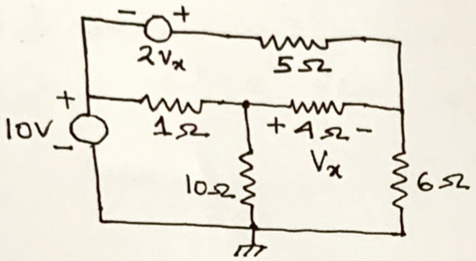


Fig. 13

(7) Find out the current 'I' in the circuit shown in fig. 14.

[Ans. I = 1.9 Amp]

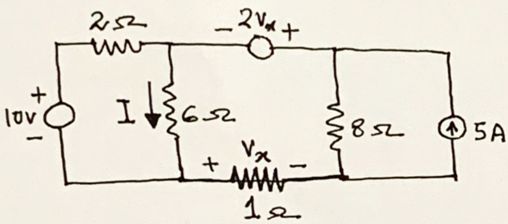


Fig. 14

(18) Determine the value of the capacitor 'C' as shown in fig. 15, for maximum power transfer to the load 10Ω.

[Ans. 50μF].

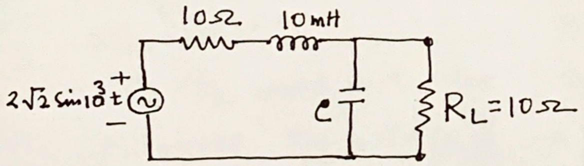


Fig. 15

(19) Find out the Rin of the circuit shown in fig. 16.

[Ans. 4.37 Ω]

[Hints: Use Δ-Y transformation]

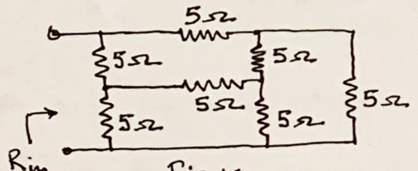


Fig. 16

(20) Determine the current through the resistance $R_L = 1\Omega$ of fig. 17.

[Ans. 0.4 A].

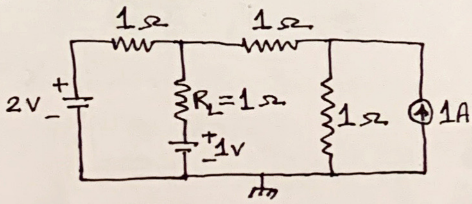


Fig. 17

ind out the value of R_L in the circuit shown in fig.18 for maximum power delivered to it.

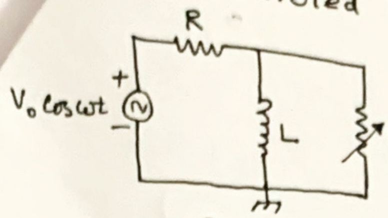


Fig. 18

[Ans. $R_L = R\omega L / \sqrt{R^2 + \omega^2 L^2}$].

(22) Find out the impedance faced by the voltage source 'V' in the circuit shown in fig.19.

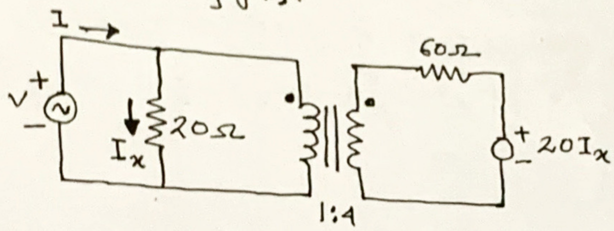


Fig. 19

[Ans. 4Ω].

(23) What will be equivalent mutual inductances of the circuits shown in fig.20(a) and fig.20(b).?

[Ans. (a) 4 H, (b) 5 H]

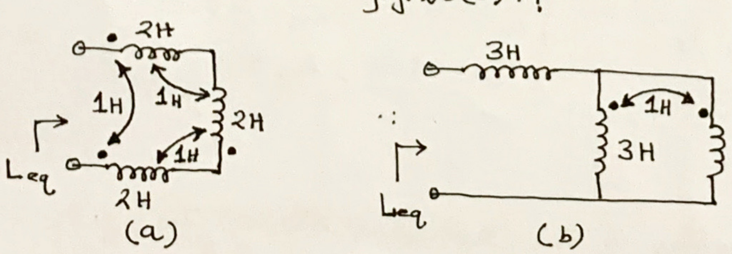
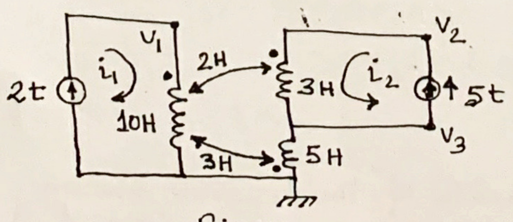


Fig. 20

(24) Determine the voltages V_1, V_2, and V_3 with respect to the ground as shown in fig. 21



[Ans. $V_1 = 30V$, $V_2 = 25V$, $V_3 = 6V$].

Fig. 21

(25) Determine the z-parameters of the network shown in fig.22. Also find out the Y-parameters using the relation between Y and Z parameters

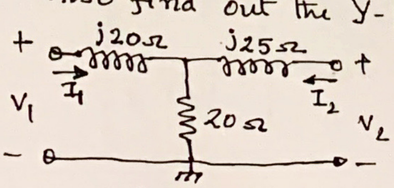


Fig. 22

[Ans. $Z_{11} = (20 + j20)\Omega$, $Z_{12} = 20\Omega$
$Z_{21} = 20\Omega$; $Z_{22} = (20 + j25)\Omega$
$Y_{11} = 0.031 \angle 112.28^\circ \Omega^{-1}$, $Y_{12} = Y_{21} = 0.019 \angle 60.94^\circ \Omega^{-1}$
$Y_{22} = 0.027 \angle 105.94^\circ \Omega^{-1}$]

26) Find out the h-parameters of the network shown in fig. 23.

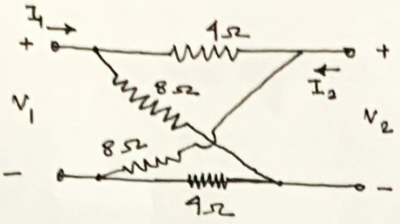


Fig. 23

$$\left[\begin{array}{ll} h_{11} = \frac{16}{3}\,\Omega \; ; & h_{12} = 1/3 \\ h_{21} = -1/3 \; ; & h_{22} = \frac{1}{6}\,\Omega^{-1} \end{array}\right]$$

27) Find out y-parameters as well as z-parameters of the network shown in fig. 2

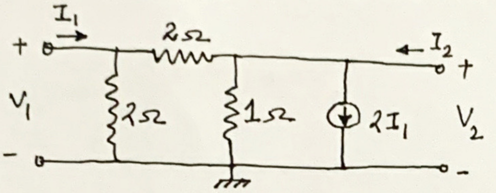


Fig. 24

[Ans. $y_{11} = 1\,\Omega^{-1}$; $y_{12} = -0.5\,\Omega^{-1}$; $y_{21} = 1.5\,\Omega^{-1}$; $y_{22} = 0.5$.
$z_{11} = \frac{2}{5}\,\Omega$; $z_{12} = \frac{2}{5}\,\Omega$; $z_{21} = -\frac{6}{5}\,\Omega$; $z_{22} = \frac{4}{5}\,\Omega$.]

28) Find out the z-parameters of the network shown in fig. 25.

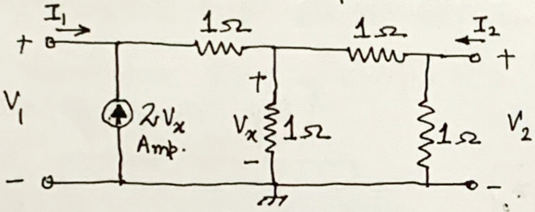


Fig. 25

[Ans.
$z_{11} = -5\,\Omega$; $z_{12} = -3\,\Omega$
$z_{21} = -1\,\Omega$; $z_{22} = 0$]

29) Determine the h-parameters of the network shown in fig. 26

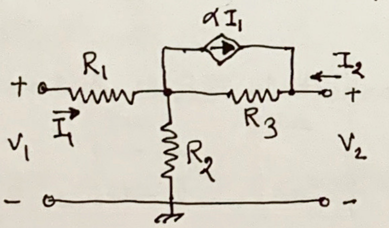


Fig. 26

[Ans. $h_{11} = R_1 + [(1-\alpha)R_2R_3/(R_2+R_3)]$
$h_{12} = R_2/(R_2+R_3)$
$h_{21} = -(\alpha R_3 + R_2)/(R_2+R_3)$
$h_{22} = 1/(R_2+R_3)$]

30) Find out the y-parameters and z-parameters of the network shown in fig. 27.

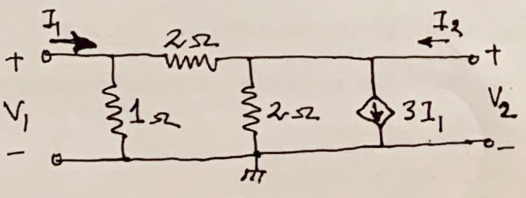


Fig. 27.

$$\begin{bmatrix} -0.4 & 0.4 \\ -3.2 & 1.2 \end{bmatrix} = [Z]$$

$$\begin{bmatrix} 1.5 & -0.5 \\ 4 & -0.5 \end{bmatrix} = [Y]$$